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The Teacher's Corner

In this section, *The American Statistician* publishes articles and notes of interest to teachers of the first mathematical statistics course and of applied statistics courses. To be suitable for this section, articles and notes should be useful to a substantial number of teachers of such a course or should have the potential for fundamentally affecting the way in which the course is taught.

Comparative Analyses of Pretest-Posttest Research Designs

DONNA R. BROGAN AND MICHAEL H. KUTNER*

Two common methods of analyzing data from a two-group pretest-posttest research design are (a) two-sample t test on the difference score between pretest and posttest and (b) repeated-measures/split-plot analysis of variance. The repeated-measures/split-plot analysis subsumes the t test analysis, although the former requires more assumptions to be satisfied. A numerical example is given to illustrate some of the equivalences of the two methods of analysis. The investigator should choose the method of analysis based on the research objective(s).

KEY WORDS: Repeated-measures/split-plot analysis; t test; Pretest-posttest designs.

1. INTRODUCTION

A common research design is the two-group pretest/posttest design with one dependent variable where subjects are not matched and may or may not be randomly assigned to the two groups (Cook and Campbell 1979). When the two groups are not formed by random assignment of subjects, a random sample from each of the two groups is necessary. This design can be extended to more than two groups; an example is the comparison of several different treatments with each other or with a control group in which each group is measured on a pretest and posttest.

The statistical analysis for these designs can be approached from several viewpoints. If the dependent variable is measured on an interval or ratio scale, a common analysis is to define a difference score for each subject (posttest minus pretest or vice versa) or a relative difference measure (the difference divided by the pretest) and then test the null hypothesis that the means or medians of the (relative) differences are equal for each group. In many cases the t test or analysis of variance is used, although nonparametric tests could also be used, for example, the Mann-Whitney U test, or the median test, or their analogs for more than two groups.

Covariance analysis, where the pretest score is used as the covariate, is another method used for analyzing this design. The difference score method is essentially a special case of the analysis of covariance where the regression coefficient of the posttest on the pretest is assumed to equal unity. Neter and Wasserman (1974, p. 717) and Cox (1958, pp. 55–56) point out that if the common slope is not near one the covariance analysis probably will be better than the difference score analysis. We note that when an experimental group is to be compared to a control group, it is often likely that inequality of slopes will prevail among groups, thus violating an assumption of the analysis of covariance. Bock (1975, Sec. 7.3) compares the interpretation of a difference-score analysis and covariance analysis and suggests guidelines regarding which analysis to use.

Still another method of analyzing this design is to view the pretest and posttest as a repeated-measures/split-plot design or as a profile of two measurements for each subject. Repeated-measures/split-plot designs are discussed in detail by Winer (1971) and Steel and Torrie (1980), whereas both repeated measures and profile analysis are discussed in Morrison (1976, Secs. 4.5, 4.6, and 5.6).

This article illustrates some of the equivalences and differences between the difference score analysis and the repeated-measures/split-plot or profile analysis. The numerical example and major discussion are for a two-group pretest/posttest design where subjects are not matched. Concluding remarks indicate how the results can be extended easily to more than two groups.

2. A NUMERICAL EXAMPLE

We consider data from Rikkers et al. (1978), who report results of a prospective randomized surgical trial allocating cirrhotic patients who had bled from varices to either a nonselective shunt (standard operation) or to a selective shunt (new operation). The dependent variable is the maximal rate of urea synthesis (MRUS), which is a quantitative test of liver function. Poor liver function is associated with a low MRUS value. MRUS was measured preoperatively and early postoperatively in eight selective shunt patients and thirteen nonselective shunt patients. The purposes of the study were to assess preoperatively the comparability of the selective and the nonselective groups and to longitudinally evaluate the change in liver function

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1. Pre and Post Maximal Rate of Urea Synthesis Level (mg urea N/hr/kg BW$^{3/4}$) and Sample Cell Means, by Group

Group	Subject	Pre	Post
Selective Shunt (new operation)	1	51	48
	2	35	55
	3	66	60
	4	40	35
	5	39	36
	6	46	43
	7	52	46
	8	42	54
Mean		$\hat{\mu}_{11} = 46.375$	$\hat{\mu}_{12} = 47.125$
Nonselective Shunt (standard operation)	9	34	16
	10	40	36
	11	34	16
	12	36	18
	13	38	32
	14	32	14
	15	44	20
	16	50	43
	17	60	45
	18	63	67
	19	50	36
	20	42	34
	21	43	32
Mean		$\hat{\mu}_{21} = 43.538$	$\hat{\mu}_{22} = 31.462$

of the two groups. Table 1 reports the MRUS values for each patient for the preoperative and postoperative periods and the respective cell means.

For completeness Table 2 displays the standard repeated-measures analysis of variance table (analysis of means method); in this example the total number of subjects, n, is 21. The hypotheses of interest to the researchers were the interaction test and the simple effects test on equality of preoperative population means between groups. The test for interaction is significant ($F = 11.36$ with 1 and 19 df, $p < .005$); therefore, it is concluded that the pre/post average change in the nonselective group is significantly different from the pre/post average change in the selective group (see the figure). In the presence of a significant interaction effect it is generally of interest to test simple effects rather than main effects (Winer 1971, p. 529). The Bonferroni multiple-comparison procedure (see Neter and Wasserman 1974) was adopted to test the following contrasts:

$$(\mu_{12} - \mu_{11}) - (\mu_{22} - \mu_{21}) = 0$$

$$\mu_{12} - \mu_{11} = 0$$

$$\mu_{22} - \mu_{21} = 0$$

and

$$\mu_{11} - \mu_{21} = 0.$$

Using an experiment wise error rate of .05, we conclude that the interaction effect is significant and μ_{21} is significantly greater than μ_{22}. Therefore, significant deterioration of liver function occurred in the nonselective patients between preoperative and early postoperative evaluation periods, whereas the selective group had no apparent deleterious effect. Two points are worth noting in the example just cited: (a) The equality of slopes test using the preoperative MRUS values as a covariate is rejected ($p < .02$); and (b) the significant interaction effect requires special handling when testing the last contrast since, for the pretest level, we have a two-group experiment in which there are no repeated measures. Therefore, the appropriate error term for this type of comparison is MS (within cell). For a more extensive coverage of this point the reader is referred to Winer (1971, pp. 529–532).

3. REPEATED-MEASURES ANALYSIS

We now discuss the statistical properties of the repeated-measures analysis of variance for this example and compare it with the statistical properties of the difference score analysis. Using the model proposed by Winer (1971, p. 519), we have

$$X_{ijk} = \mu + \alpha_i + \Pi_{k(i)} + \beta_j + \alpha\beta_{ij} + \beta\Pi_{jk(i)} + \epsilon_{m(ijk)} \quad (3.1)$$

$j = 1, 2$ (pretest = 1, posttest = 2),

$i = 1, 2$ (group 1 = 1, group 2 = 2),

$k = 1, 2, \ldots, n_i, \quad m = 1,$

where X_{ijk} is the observed value of subject k within group i at time j,

μ is the overall mean,
α_i is the effect of group i,
$\Pi_{k(i)}$ is the effect of subject k nested within group i,
β_j is the effect of the repeated-measures variable j (*i.e.*, pretest and posttest),
$\alpha\beta_{ij}$ is the interaction of group i with level j of the repeated measures factor,

2. Repeated-Measures Analysis of Variance for Maximal Rate of Urea Synthesis Level

Source of Variation	df	Sum of Squares	Mean Squares	F Ratio
Between Subjects	20 (n − 1)			
Groups	1	847.48	847.48 (MS_G)	3.63 (MS_G/MS_E)
Subjects Within Groups	19 (n − 2)	4440.00	233.68 (MS_E)	
Within Groups	21 (n)			
Pre/Post	1	317.69	317.69 (MS_P)	8.86 (MS_P/MS_{PE})
Groups x Pre/Post	1	407.41	407.4 (MS_{GP})	11.36 (MS_{GP}/MS_{PE})
(Pre/Post) x Subjects Within Groups	19 (n − 2)	681.21	35.85 (MS_{PE})	

and

$\beta\Pi_{jk(i)}$ is the interaction of subject k within group i with level j of the repeated-measures factor.

The following constraints are imposed on the parameters:

$$\alpha_{\cdot} = \beta_{\cdot} = \alpha\beta_{i\cdot} = \alpha\beta_{\cdot j} = 0,$$

where

$$\alpha\beta_{\cdot j} = \sum_i \alpha\beta_{ij}, \text{ and so on.} \tag{3.2}$$

In the design under discussion, the repeated-measures factor and the group factor are each at two levels.

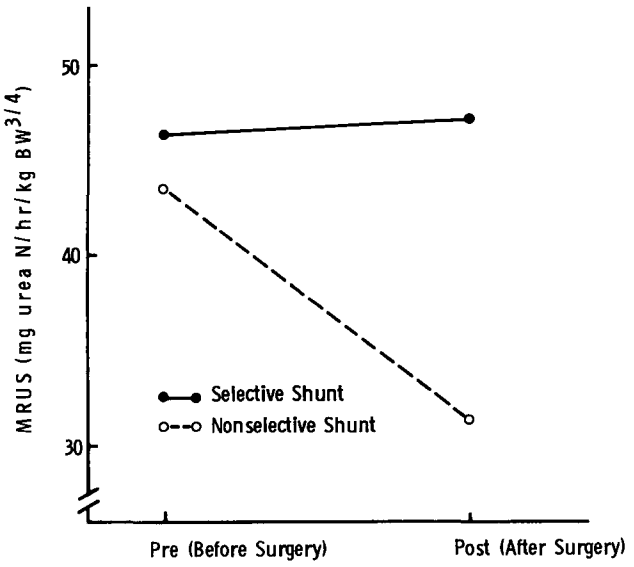
The general analysis of variance also is indicated in Table 2, where n is the total number of subjects. Note that it is not necessary for each group to contain the same number of subjects. Assuming the group factor and the pre/post factor to be fixed effects, Winer (1971) shows that the appropriate F tests are as indicated in the F ratio column of Table 2.

It is worth noting exactly what null hypotheses are tested in Table 2. The ratio MS_G/MS_E tests the null hypothesis that there is no *group main effect*. This is equivalent to testing whether the *sum* of the pretest and posttest observations on each subject has the same population mean in the two groups. The ratio MS_P/MS_{PE} tests the null hypothesis that there is no *pre/post main effect* and is equivalent to testing whether the population mean of the pretest observations is the same as the population mean of the posttest observations. The ratio MS_{GP}/MS_{PE} tests the null hypothesis that there is no *interaction* between the group main effect and the pre/post main effect. This ratio also tests whether the difference between pretest and posttest observations has the same population mean in both groups. This is the test many researchers are interested in when using this research design, since they often wish to assess whether a treatment has had any effect upon an experimental group. Note that this F test has $(1, n - 2)$ df, which will correspond to the t test with $(n - 2)$ df, as discussed in the next section.

Two assumptions are required to arrive at the F tests indicated in Table 2 (Winer 1971).

1. The pretest and posttest population variance-covariance matrices for each group are assumed equal.
2. The random effects $\Pi_{k(i)}$, $\beta\Pi_{jk(i)}$, and $\epsilon_{m(ijk)}$ from the model in (3.1) are all independently and normally distributed with mean zero and variances σ_Π^2, $\sigma_{\beta\Pi}^2$, and σ_E^2, respectively.

Assumption (1), equality of the variance-covariance matrices, implies two other results worth noting. First, the variation of subjects within the two groups is homogeneous. That is, if each subject's pretest and posttest observations are added together, this sum has the same population variance in both groups. This allows pooling over groups to calculate SS_E. Second, the variation of the interaction of subject and the pre/post factor is homogeneous for the two groups. That is, if the difference score between pretest and posttest is defined for each subject, the population variance of the difference scores is the same for both groups. This allows pooling over groups to calculate SS_{PE}.



Mean Pre and Post Maximal Rate of Urea Synthesis Level (MRUS) by Type of Surgery

4. DIFFERENCE-SCORE ANALYSIS

Using model (3.1) and forming a difference score d_{ik} for each subject k nested in group i yields

$$\begin{aligned} d_{ik} &= X_{i1k\cdot} - X_{i2k} \\ &= (\beta_1 - \beta_2) + (\alpha\beta_{i1} - \alpha\beta_{i2}) + (\beta\Pi_{1k(i)} - \beta\Pi_{2k(i)}) \\ &\quad + (\epsilon_{m(i1k)} - \epsilon_{m(i2k)}). \end{aligned} \tag{4.1}$$

The term $(\alpha\beta_{i1} - \alpha\beta_{i2})$ is a parameter associated with group i and measures the "effect" of group i on the difference score d_{ik}. The null hypothesis to be tested is H_o: $\alpha\beta_{11} - \alpha\beta_{12} = \alpha\beta_{21} - \alpha\beta_{22}$ or H_o: $\alpha\beta_{11} - \alpha\beta_{12} = \alpha\beta_{21} - \alpha\beta_{22} = 0$. The difference scores in (4.1) can be viewed as a one-way classification model in which the error term is the sum of the following two terms:

$$(\beta\Pi_{1k(i)} - \beta\Pi_{2k(i)}) \quad \text{and} \quad (\epsilon_{m(i1k)} - \epsilon_{m(i2k)}).$$

If we assume this error term and also homogeneous variances for the two groups, an appropriate test statistic is the Student t test for two independent samples with $(n - 2)$ df.

5. COMPARISON OF THE TWO ANALYSES

The following three results are useful computationally and can be verified easily with the example from Table 1.

1. If the sum of the pretest and posttest is formed for each subject and a two-sample t test is used to compare the group means of the sum, then the calculated $t = 1.904$ and is the square root of the F test for Groups in Table 2 with (1,19) df.

2. If the difference between the pretest and the posttest is formed for each subject and a two-sample t test is used to compare the group means of these differences, then the calculated $t = 3.371$ and is the square root of the F test for Groups x (pre/post) in Table 2 with (1,19) df.
3. If all 21 subjects are considered to be in one group, then the t statistic to test the null hypothesis that the mean difference score is zero has 20 df and equals 3.158. From Table 2, if we reanalyze the within-subjects component by assuming that the Group x (pre/post) interaction is zero and SS_{GP} is pooled with SS_{PE}, then the F test for the main effect pre/post yields $F = 9.97$ with (1,20) df, which equals the square of the preceding t statistic.

These results demonstrate that the various F tests in the repeated-measures analysis of variance can be obtained by using simple t tests on linear combinations of the pre and post scores. It can be shown algebraically that interpretations (1), (2), and (3) of the F tests hold in the particular research design discussed in this article, that is, a two-group pretest/posttest design. In fact, the numerical operations in (1) and (2) of summing and differencing the pretest/posttest observations are used by the latest version of the BMD P2V program in calculating sums of squares in repeated-measures designs (Dixon and Brown 1979).

Since the difference-score analysis is embedded in the repeated-measures analysis, the repeated-measures analysis provides more information about the data at hand. Fewer assumptions, however, are required in the difference-score analysis. The difference-score analysis assumes only homogeneous variances for the difference scores and a normally distributed error term with mean zero. It is easy to show that if the assumptions of the repeated-measures analysis are satisfied, then the assumptions of the difference-score analysis are also met. However, the converse is not true.

In our experience, the researcher rarely is interested in only the interaction test, that is, the difference-score analysis. Furthermore, simple effects are commonly of interest even in the no interaction effect experiments. Therefore, we advocate the use of the repeated-measures/split-plot analysis in most instances. However, we urge the user to empirically validate the underlying assumptions.

In the example discussed in Section 2, the researchers would have been interested in assessing the significance of the main effect time (pre vs. post) *if* the Groups x pre/post interaction had been nonsignificant. That is, the nonsignificant interaction would have indicated that the two groups did not differ significantly on their MRUS difference scores. The pre/post main effect test would then indicate whether the MRUS difference score in both groups was significantly different from zero, that is, whether the treatment did or did not effect both groups.

6. GENERALIZATION OF FINDINGS

If there are more than two groups, similar results can be obtained. The difference-score analysis would no longer be performed by a t test but by an F test in a one-way analysis of variance. A further extension can be made where the different groups being compared may be defined by several factors in a factorial design.

However, there is no logical extension of this discussion to more than two levels of the repeated-measures factor since a simple difference score analysis would no longer be appropriate.

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